

A Complete Algorithmic Toolbox for Congressional Redistricting: Legal Requirements, Algorithm Components, and Callais-Compliant Design for the Post-2026 Redistricting Cycle

B.0 Series Synthesis

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Abstract

Congressional redistricting must satisfy eleven distinct legal requirements spanning constitutional population equality, geographic compactness, county preservation, the Voting Rights Act, and the racial-predominance bar set by *Shaw v. Reno* (1993). The 2026 Supreme Court decision in *Louisiana v. Callais* adds a critical constraint: race-conscious and partisan signals must be kept *disentangled* in any redistricting algorithm—they cannot be combined in a single optimisation run.

We present a unified algorithmic toolbox organised along three axes: (A) vertex weights governing what METIS balances, (B) edge management controlling compactness and subdivision preservation, and (C) division strategy determining how the bisection tree is structured. Fifteen algorithm components developed across the B-series papers (B.1–T.8) map to these axes, and each legal requirement maps to at least one component.

The toolbox is structured so that each mode uses exactly one signal type. **VRA-Section** (T.7) addresses Gingles Prong 1 using geographic alignment alone—no racial target, no racial count. **ProportionalSection** (T.5, planned) addresses proportionality using partisan vote share alone. A `callais_preflight` gate enforces mutual exclusion: the two signals cannot be combined in one run. This is not a limitation but the legally required architecture under *Louisiana v. Callais*.

We present a bakeoff of eight algorithm combinations across four legally contested states (Wisconsin 8D, North Carolina 14D, Georgia 14D, Alabama 7D), measuring edge cut, population balance, D-seat count, majority-minority district count, natural bisection ratio, and county split count. The tables intentionally retain a mixture of confirmed, estimated, and pending cells and therefore identify hypotheses rather

than a completed comparative experiment. They illustrate that geography and algorithm choice can both affect partisan and minority-opportunity diagnostics; they do not establish causation, VRA compliance, or a legally required algorithm. A separately preregistered Wisconsin tract-level proof slice retained an initial two-pass/two-fail result that exposed tolerance, reporting, and contiguity-path defects. After remediation, all four neutral structure runs passed native audits and regenerated exactly. This is a one-input implementation result, not an algorithm ranking.

For prospective bakeoffs we add a second, non-combinable procedural scoreboard. It records whether an arm is replay-only, audit-packaged, tree-certified, or released through a certified public ceremony; it also records cut-proof coverage, prefix verification, completion or halt status, transcript size, verifier time, and hostile-test results. Existing frozen results are not retroactively upgraded, and no scalar score blends these fields with map quality.

A court can audit exactly which signals were used in any plan by reading the `whatif-manifest v1` JSON sidecar produced by the `bisect` CLI. Every run is parameter-reproducible and hash-chain verifiable. Certified ceremony runs additionally commit the rule before the root cut and release later depths only after frozen public-review intervals.

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1 Introduction

1.1 The Redistricting Problem

Every ten years, following the decennial census, all 50 American states must redraw their congressional district boundaries. The process allocates 435 seats across states via the Huntington-Hill method (Article I, §2, operationalised since 1941), then asks each state to partition its population into exactly k districts where k is its apportioned seat count. The constitutional floor is simple: districts must be as close to equal in population as practicable (*Wesberry v. Sanders*, 1964, one person one vote). Everything above that floor has accumulated as a dense forest of legal requirements over six decades of redistricting litigation.

The forest has grown denser since 2020. *Rucho v. Common Cause v. Common Cause* (2019) removed federal courts from partisan gerrymandering review, but *immediately redirected challengers to state courts* under state constitutional grounds. Pennsylvania’s *League of Women Voters* (2018) and North Carolina’s *Harper v. Hall* (2022) established that state constitutions can ban partisan gerrymandering even where *Rucho* closes the federal door. *Allen v. Milligan* (2023) reaffirmed and clarified the Voting Rights Act §2 obligation to draw minority-opportunity districts where Gingles conditions are met. And *Louisiana v. Callais* (2026) settled the disentanglement question: a state may not use race-conscious criteria to justify what is actually partisan manipulation, and the “strong-inference” framework requires challengers to show that a state’s stated political goals could have been achieved *with better minority outcomes*—meaning the algorithm must be capable of separating the two signals.

This paper provides what the litigation landscape demands: a *toolbox* of algorithm components where each tool addresses exactly one legal requirement, and the tools compose without conflict because they are legally required to be separate.

1.2 Why a Toolbox, Not a Single Algorithm

No single redistricting algorithm can satisfy all requirements simultaneously. The requirements pull in different directions. Geographic compactness (round districts) conflicts with county preservation (keeping existing political lines). VRA minority-opportunity district formation requires concentrating minority population, which can conflict with equal geographic spread. Proportionality requires a partisan signal, which *Louisiana v. Callais* prohibits from being mixed with a racial signal. Census stability across decades requires that the algorithm converges on the same geographic structure regardless of which census year’s data is used.

The toolbox architecture acknowledges these tensions explicitly. Each tool is a parameter dial: the *geographic edge weights* of B.2 control compactness; the *county-sticky weights* of T.3 control subdivision preservation; the *VRASection alignment score* of T.7 controls Gingles Prong 1 alignment; the *AreaSection dual constraint* of T.2 controls geographic fairness. These dials can be tuned independently. They cannot all be maximised simultaneously, but the tradeoffs are *explicit and auditable*: a court can read the parameter values from the plan manifest and understand exactly which tradeoffs were made.

1.3 The Callais Constraint

Louisiana v. Callais, 608 U.S. ___ (2026), has three holdings relevant to algorithmic redistricting.

First, Louisiana’s extra minority district was unconstitutional racial gerrymandering: the VRA did not require a second majority-minority district where the first compactness prong of *Thornburg v. Gingles* (1986) was not met.

Second, §2 challengers must use the “strong-inference” framework: they must show that the state’s *stated political goals* (partisan advantage, community preservation, compactness) could have been achieved *with better minority outcomes*. This shifts the burden from “the map is racially unfair” to “a race-neutral algorithm could have done better on both dimensions”. The toolbox is precisely positioned to provide this counterfactual: one can run GeoSection (no minority signal) and VRASection (geographic minority alignment) on the same state and compare.

Third, at p. 36, the majority holds that race-conscious and partisan signals must be *disentangled*: a redistricting run that mixes minority VAP concentrations as a constraint alongside partisan vote share as a constraint is constitutionally suspect because a court cannot separate the racial motivation from the partisan motivation in the resulting map.

This third holding is the architecturally decisive constraint. It means that a redistricting system with multiple simultaneous signals— minority population fractions plus partisan vote shares in the same METIS objective function—fails the Callais disentanglement test. The toolbox’s structure, where each mode uses exactly one signal type, is therefore not a simplification or limitation but a constitutionally correct design.

1.4 Paper Scope and Contributions

This paper is the capstone of the B-series algorithmic redistricting programme. It synthesises findings from B.1 through T.8 into a unified framework.

Contributions.

1. A complete mapping of all eleven legal redistricting requirements to specific algorithm components (Section 4).
2. A three-axis taxonomy of algorithm variants (Section 3).
3. A bakeoff of eight algorithm combinations across four litigated states, with empirical results on compactness, partisanship, minority representation, and county preservation (Section 5).
4. A formal Callais compliance architecture showing how the toolbox enforces p. 36 disentanglement at the CLI level (Section 6).
5. Practitioner recommendations for which configuration to choose for which legal context (Section 7).

Notation. Throughout this paper, $G = (V, E, w)$ denotes the state’s census-tract adjacency graph with vertex population weights and edge boundary-length weights. k denotes the state’s congressional seat count. $\text{EC}(i : j)$ denotes the minimum edge-cut at split ratio $i : j$. $\text{EC}_{\text{norm}}(i : j) = \text{EC}(i : j) / \sqrt{\min(i, j)}$ denotes the isoperimetrically normalised edge-cut introduced in GeoSection (T.1). $p^* = L(0.5)$ denotes the Lorenz-curve natural population fraction (population in the densest 50 % of land area). $\text{D votes \%}$ is the 2020 presidential Democratic two-party vote share.

2 Legal Requirements for Congressional Redistricting

Congressional redistricting must satisfy a hierarchy of legal requirements derived from the federal Constitution, the Voting Rights Act, and state constitutions and statutes. We enumerate eleven requirements that any redistricting algorithm must address, organising them from the hardest federal floor to the softer state-specific ceilings. Each requirement has a precise algorithmic interpretation that Section 4 will map to specific toolbox components.

2.1 Constitutional Floor Requirements

R1: Population Equality ($\pm 0.5\%$)

Legal basis: *Wesberry v. Sanders* v. Sanders, 376 U.S. 1 (1964) (“one person, one vote”); *Karcher v. Daggett*, 462 U.S. 725 (1983) (near-perfect equality required for congressional districts).

Congressional districts must contain as close to equal population as mathematically possible. Unlike state legislative districts, which allow deviations up to $\pm 10\%$ under *Mahan v. Howell* (1973), congressional districts face a near-zero tolerance: any deviation must be “unavoidable” or justified by a legitimate state policy.

Algorithmic interpretation: All METIS calls use $\text{ubvec}[0] = 1.001$ (0.1 % tolerance per part, yielding $\leq 0.5\%$ maximum deviation across k parts by the triangle inequality). Population balance is the hard constraint; all other objectives are soft.

R2: Contiguity

Legal basis: Article I, §2 (no explicit text, but universally treated as a constitutional floor by courts); all 50 state constitutions require contiguous districts.

Every district must be a connected geographic region. No district may consist of non-adjacent parcels even if they share the same population.

Algorithmic interpretation: METIS’s `-contig` flag is set at every bisection level. For planar census-tract adjacency graphs, METIS empirically produces contiguous partitions even without this flag [Della-Libera, 2026b], but the flag is set explicitly to provide a formal guarantee.

2.2 Compactness Requirements

R3: Compactness

Legal basis: Most state constitutions (California Constitution Art. XXI §2(d); New York ELEC §4-100; Iowa Code §42.4); some states treat compactness as a federal equal-protection requirement under *Shaw v. Reno v. Reno*, 509 U.S. 630 (1993) (“bizarre” shapes trigger heightened scrutiny).

Districts should be reasonably compact in shape: not elongated, not fractal, not reaching tentacles into remote areas to capture or avoid specific communities. Courts have measured compactness via the Polsby-Popper score (ratio of district area to area of circumscribed circle), the Reock score (ratio to minimum bounding circle), and the isoperimetric ratio (perimeter squared over area).

Algorithmic interpretation: Geographic edge weights (B.2) encode TIGER boundary lengths as edge costs, making it expensive for METIS to cut across long geographic boundaries. The isoperimetric normalisation of GeoSection (T.1) additionally prevents the caterpillar pathology (sequential urban peeling) that creates compactness violations invisible to per-district metrics but visible in the bisection tree structure.

2.3 Subdivision Preservation

R4: County and Subdivision Preservation

Legal basis: *Rucho v. Common Cause*, 588 U.S. 684 (2019) (partisan gerrymandering nonjusticiable, but traditional criteria—including county preservation—remain valid state-court grounds); *Harper v. Hall* (N.C. Sup. Ct., 2022) (required county grouping); most state redistricting statutes list “preserving political subdivisions” as an explicit criterion.

Districts should avoid splitting counties, municipalities, and townships where possible, consistent with the population-equality requirement. County preservation has two rationales: it minimises disruption to existing community administrative structures, and it provides a measurable baseline that prevents “county-splitting gerrymandering” where a partisan actor systematically splits unfriendly counties to dilute their political influence.

Algorithmic interpretation: County-sticky edge weights (T.3) multiply edge costs by $(1 + \alpha_{\text{county}})$ for pairs of census tracts sharing the same county FIPS code. As $\alpha_{\text{county}} \rightarrow \infty$, county splits occur only when the population-equality constraint forces them. The α parameter is a published number; a court can read it from the plan manifest.

2.4 Voting Rights Act Requirements

R5: VRA §2 Gingles Prong 1 — Minority Geographic Compactness

Legal basis: 52 U.S.C. §10301; *Thornburg v. Gingles*, 478 U.S. 30 (1986) (three-prong test: minority group sufficiently large and geographically compact; minority cohesion; majority bloc voting); *Allen v. Milligan*, 598 U.S. 1 (2023) (Alabama required two Black-opportunity districts); *Louisiana v. Callais*, 608 U.S. ___ (2026) (Gingles Prong 1 is geographic, not racial targeting).

§2 requires states to draw majority-minority districts where a minority community is (1) sufficiently large and geographically compact to form a majority in a single district, (2) politically cohesive, and (3) faces bloc voting by the majority. The geographic concentration prong (Prong 1) is the one that algorithmic redistricting can directly address: the algorithm must be capable of identifying whether a compact minority community exists at the geographic scale of a congressional district.

Algorithmic interpretation: VRASection (T.7) adds a geographic alignment score $A = |\text{MVAP}_{\text{frac}}(L) - \text{MVAP}_{\text{frac}}(R)|$ to the ratio selection objective. This score is a *geographic property*: it measures whether the bisection places minority population on one side. Critically, the score uses only the spatial distribution of minority VAP, not racial identity or any racial target. At $w_{\text{vra}} = 0.40$, compactness is weighted 50 % more heavily than alignment, ensuring racial geometry does not predominate.

R6: No Racial Predominance

Legal basis: *Shaw v. Reno* v. Reno, 509 U.S. 630 (1993); *Miller v. Johnson* v. Johnson, 515 U.S. 900 (1995) (race cannot be the “predominant factor” in drawing district lines); *Louisiana v. Callais* p. 36 (race-conscious and partisan signals must be disentangled).

A district whose shape is “so bizarre” that it can only be explained by race triggers strict scrutiny under *Shaw v. Reno*. Under *Miller v. Johnson*, any time race is the predominant factor overriding traditional criteria (compactness, contiguity, political subdivisions), the district requires a compelling interest.

Algorithmic interpretation: VRASection’s $w_{\text{vra}} = 0.40$ weight is calibrated to fall well below the *Shaw v. Reno*–*Miller v. Johnson* racial-predominance threshold. The score does not specify a racial target (no “district must be 50 % Black VAP”); it uses only the geographic alignment of existing population distribution. The design ensures that compactness remains the predominant objective (see Section 6 for the formal analysis).

2.5 Partisan Neutrality

R7: Partisan Neutrality (State Courts)

Legal basis: *League of Women Voters v. Commonwealth of Pennsylvania* (Pa. Sup. Ct., 2018) (partisan gerrymander violates Pennsylvania Constitution); *Harper v. Hall* (N.C. Sup. Ct., 2022) (initial decision found unconstitutional partisan gerrymander); *Harkenrider v. Hochul* (N.Y. Ct. App., 2022) (struck down partisan gerrymander under New York Constitution).

While *Rucho v. Common Cause* (2019) closed the federal door to partisan gerrymandering claims, state courts in Pennsylvania, North Carolina, New York, and other states have entertained partisan-neutrality claims under state constitutional grounds. The legal standard varies: some courts require proportionality of seats to votes, others require map-making processes that do not give undue advantage.

Algorithmic interpretation: The Rodden-null test is the key tool here. The T.2 Area-Section paper demonstrated that AreaSection and GeoSection produce identical seat counts in 76 % of states and differ by only 1 seat in the remaining states, with no systematic partisan

direction. The partisan outcomes in all competitive states are explained by geographic voter sorting (the “Rodden effect” [Rodden, 2019]), not by algorithmic bias. ProportionalSection (T.5, planned) addresses this directly by incorporating partisan vote share as a separate, Callais-compliant mode.

2.6 Cross-Census Stability

R8: Census Stability (Litigation Argument)

Legal basis: Not a statutory requirement, but a powerful litigation argument against partisan gerrymandering claims. If a redistricting algorithm produces the same map—the same natural bisection ratio, the same seat counts—across 2000, 2010, and 2020 Census data, then the map is not a product of the current census cycle’s partisan distribution. A challenger cannot argue the algorithm exploited the current population counts to achieve partisan ends when the identical map would have been drawn from twenty-year-old data.

Algorithmic interpretation: StabilitySection (T.8) measures the Census Stability Score (CSS): a composite of seat-count stability, ratio stability, and proportionality-gap stability across census years. The 2000–2010 sweep found 67% of states are stable at the $\Delta f \leq 0.05$ threshold; Iowa is the canonical unstable state. A $CSS \geq 0.67$ threshold is proposed as the minimum for making the census-stability legal argument.

2.7 Auditability and Seed Invariance

R9: Seed and Auditability Invariance

Legal basis: No explicit legal requirement, but a pervasive requirement in redistricting litigation: courts need to be able to reproduce the map from the stated parameters. A plan produced by an algorithm whose outputs vary with undisclosed random seeds is not auditable; a challenger can always claim the seed was chosen to achieve a specific partisan outcome.

Algorithmic interpretation: The B.7 paper certified that GeoSection achieves near-zero partisan variance across random seeds (seat-count variance < 0.3 seats per state at 50 seeds per ratio). ApportionRegions (T.4, planned) extends this to the Huntington-Hill linkage: the natural bisection ratio can be derived from the state’s prime factorisation of district counts, making it parameter-free and auditable. The `bisect` CLI records every parameter in the `whatif-manifest v1` JSON sidecar, and a hash chain verifies that no parameter was altered after the run.

2.8 Multi-Chamber Consistency

R10: Multi-Chamber Consistency

Legal basis: State constitutions require that congressional and state legislative district maps be drawn consistently; the “one-map” approach prevents “stacked” gerrymandering where a partisan actor controls all three chambers and packs or cracks a community at every level.

Algorithmic interpretation: NestSection (T.6) forces all chambers sharing a state to use a *compatible factorisation spine* derived from $g = \gcd(C, S, H)$ where C , S , H are the congressional, senate, and house seat counts. Eleven states achieve exact nesting (spine compatibility score 0); the remainder require best-effort Mode 3 nesting. A shared spine is an anti-gerrymandering mechanism: a partisan actor who locks in a congressional tree cannot then draw senate boundaries that contradict it.

2.9 Huntington-Hill Linkage

R11: Apportionment Linkage

Legal basis: Article I, §2 (apportionment proportional to population); the Huntington-Hill method (2 U.S.C. §2a) as the statutory apportionment algorithm. A redistricting algorithm that uses a bisection tree structurally linked to the state’s Huntington-Hill seat count provides an additional layer of constitutional grounding: the geographic partition reflects the same proportionality principle as the apportionment itself.

Algorithmic interpretation: ApportionRegions (T.4, planned) determines the bisection tree structure from the prime factorisation of k , mirroring the successive subdivision implicit in the Huntington-Hill priority sequence. A state with $k = 8 = 2^3$ can be exactly bisected three levels deep. A state with $k = 7$ (prime) requires a near-equal first split and defers the odd seat to the final level. The linkage is not a hard constitutional requirement but provides additional grounding for the algorithm’s objectivity claim in litigation.

3 The Algorithm Toolbox: Three Axes

The B-series papers collectively define fifteen algorithm components, organised along three orthogonal axes. Each axis is a dial that can be set independently, and the legal requirements in Section 2 map cleanly to at most one dial per requirement. This orthogonality is the key property that makes the Callais disentanglement constraint satisfiable: the race dial (VRASection) and the partisan dial (ProportionalSection) are on different axes and can never be set simultaneously without triggering the `callais_preflight` gate.

Figure 1 provides a visual overview of the three axes and their components.

3.1 Axis A: Vertex Weights (What METIS Balances)

METIS’s graph bisection algorithm balances one or two “constraints” per vertex, specified as integer weight vectors. The choice of what to balance is the first algorithmic design decision.

A1: Population Only ($ncon=1$)

The baseline choice: each vertex carries one weight (population from the census), and METIS balances population fractions. This is the standard used in B.1 (recursive bisection) and B.7 (solution space certification). It satisfies R1 (population equality) and is neutral with respect to all other requirements.

METIS configuration: $ncon = 1$, $ubvec[0] = 1.001$.

Three-Axis Algorithm Taxonomy		
Axis A: Vertex weights What METIS balances	Axis B: Edge weights Boundary cost function	Axis C: Division strategy How the bisection tree is structured
<i>ncon=1, pop only</i> Standard (B.1, B.7, T.1)	<i>Unweighted</i> Minimal intervention	<i>Balanced bisection</i> Standard $k/2 : k/2$
<i>ncon=2, pop+area</i> AreaSection (T.2)	<i>Geographic (TIGER)</i> Compactness	<i>GeoSection (T.1)</i> Isoperimetric ratio scan
<i>ncon=1+alignment</i> VRASection (T.7)	<i>County-sticky (α)</i> Subdivision preservation	<i>AreaSection (T.2)</i> Dual-constraint ratio scan
<i>ncon=2, pop+D_votes</i> ProportionalSection (T.5)		<i>ApportionRegions (T.4)</i> Prime factorisation tree
		<i>NestSection (T.6)</i> Compatible spine
		<i>StabilitySection (T.8)</i> Cross-census CSS filter

Figure 1: Three-axis taxonomy of the algorithmic redistricting toolbox. Axis A controls what METIS optimises (population fractions to balance). Axis B controls how edges are weighted (which boundaries are expensive to cut). Axis C controls how the bisection tree is structured. Each legal requirement maps to at most one component on one axis.

A2: Population + Land Area (*ncon=2*) — AreaSection

AreaSection (T.2) adds a second METIS constraint: land area. Each vertex carries two weights (population, area in hectares). METIS simultaneously targets 50 % population balance and 50 % area balance at the first bisection level, with the population constraint taking priority when the two conflict.

The area constraint eliminates *urban peeling*—the systematic first-level separation of compact metropolitan cores from the rural remainder. Urban peeling is not a constitutional violation but creates implicitly packed maps even with zero partisan input: every recursion level concentrates Democratic voters into one compact district. Area balance prevents this by requiring that both halves of the state contain approximately equal land areas, forcing the algorithm to cross urban-rural geographic boundaries.

METIS configuration: $ncon = 2$, $ubvec = [1.001, 1.10]$, $tpwgts = [p_L, 1 - p_L, 0.5, 0.5]$ for population fraction p_L at ratio $p_L : 1 - p_L$. The flat C-array is row-major over constraints: entries 0–1 are the part-weights for constraint 0 (population), and entries 2–3 are

the part-weights for constraint 1 (area). That is, the layout is $[\text{pop}_L, \text{pop}_R, \text{area}_L, \text{area}_R]$, *not* the interleaved form $[\text{pop}_L, \text{area}_L, \text{pop}_R, \text{area}_R]$ which would silently mis-weight both constraints.

Rodden-null result (T.2): AreaSection produces identical seat counts to GeoSection in 76 % of states (25/34 with complete data). We exclude the 7 single-district states ($k = 1$: Alaska, Delaware, Montana, North Dakota, South Dakota, Vermont, Wyoming) and 9 states where the algorithm produces identical partisan outcomes under all tested modes, leaving 34 states where algorithm selection affects partisan outcomes; the 76 % figure is computed over this contested subset. The 9 divergent states differ by exactly 1 seat, with no systematic partisan direction. The mean proportionality gap in competitive states is -6.2 pp under both algorithms, confirming geographic voter sorting as the dominant effect.

A3: Population + Alignment Signal — VRASection

VRASection (T.7) does not add a second METIS constraint. Instead, it modifies the *ratio selection objective* at each bisection level by adding a geographic alignment score:

$$\text{score}(\text{ratio}) = \text{EC}_{\text{norm}} - w_{\text{vra}} \cdot A \cdot \max(\text{EC}_{\text{norm}}, 1)$$

where $A = |\text{MVAP}_{\text{frac}}(L) - \text{MVAP}_{\text{frac}}(R)|$ is the alignment score (how much minority population is concentrated on one side).

This design explicitly avoids the MetisVra failure mode [Della-Libera, 2026a]: minority VAP fractions (0.0–1.0) and census population counts (10,000–200,000) differ by $60\text{--}800\times$, causing numerical degeneration in any METIS multi-constraint formulation that combines them. VRASection bypasses this by using A only as a *ratio selector*, not as a METIS constraint.

Key property (Callais compliance): The alignment score A uses only the spatial distribution of minority VAP, not a racial target. At $w_{\text{vra}} = 0.40$, compactness is weighted 50 % more heavily than alignment. The algorithm cannot see how many people of any race are in any district; it sees only whether one side of a proposed bisection has more minority VAP *concentration* than the other.

A4: Population + Partisan Votes ($n_{\text{con}}=2$) — ProportionalSection

ProportionalSection (T.5, planned) is the *Callais-mutex partner* of VRASection. It adds partisan vote share (D-vote fraction per tract) as a second METIS balance constraint, targeting proportional representation.

Callais constraint: ProportionalSection is mutually exclusive with VRASection. The `callais_preflight` gate in the `bisect` CLI enforces this mutex: any attempt to run both signals simultaneously is rejected with an explicit [BOUNDARY] error citing *Louisiana v. Callais* p. 36.

3.2 Axis B: Edge Management

METIS minimises the total cost of cut edges. Edge weights determine which boundaries are expensive to cut, thereby shaping the resulting district geometry.

B1: Unweighted Edges

All edges have weight 1 (or boundary length in metres from TIGER, but not amplified). This is the baseline of Paper B.1. METIS minimises the number of cut adjacency relationships, producing districts that respect geographic adjacency but do not specifically favour compact shapes.

B2: Geographic Edge Weights (TIGER Boundary Lengths)

Paper B.2 (edge-weighted bisection) demonstrated that weighting edges by their TIGER boundary length produces a 56 % improvement in mean Polsby-Popper compactness over unweighted bisection. The mechanism is direct: long borders (between adjacent tracts with extensive shared boundary) are expensive to cut, so METIS prefers to keep them intact. Districts therefore tend to follow natural geographic contours (rivers, ridgelines, coastlines) rather than cutting across them.

Compactness result (B.2): Mean Polsby-Popper improves from $PP = 0.25$ (unweighted) to $PP = 0.39$ at standard geographic weighting, exceeding the mean Polsby-Popper of 2020 enacted congressional maps ($PP = 0.32$).

B3: County-Sticky Weights — SubdivisionBisect

Paper T.3 introduces the governmental stickiness weight:

$$w'(u, v) = w_{\text{geo}}(u, v) \times (1 + \alpha_{\text{county}} \cdot \mathbf{1}[\text{county}(u) = \text{county}(v)])$$

Pairs of tracts sharing the same county FIPS code are harder to separate by a factor of $(1 + \alpha_{\text{county}})$. As α_{county} increases, county splits are reduced, approached only when population-equality arithmetic forces them. The formula generalises to municipality, township, and precinct levels via α_{mcd} , α_{place} , α_{vtd} .

Legal hook: “Preserving political subdivisions” is an explicit statutory criterion in most state redistricting codes. T.3 provides the first formal algorithmic operationalisation of this criterion as a tunable parameter.

3.3 Axis C: Division Strategy

The division strategy determines how the recursive bisection tree is structured: what split ratios are tried, how ratios are selected, and how the tree is linked to other legal constraints.

C1: Balanced Bisection (Standard)

The standard approach: always split at $\lfloor k/2 \rfloor : \lceil k/2 \rceil$. This is the baseline of B.1 (recursive bisection) and B.7 (solution space). It produces a perfectly balanced binary tree, which is computationally efficient and well-studied. Its weakness is the caterpillar pathology: in states with compact urban cores, the balanced split at each level *still* costs less than any alternative because the urban boundary is shorter than any geographic bisection of the full state—even after the 1:(k-1) ratio has already been excluded from consideration.

C2: GeoSection — Isoperimetric Ratio Scan

GeoSection (T.1) scans all feasible split ratios $1 : (k - 1)$ through $\lfloor k/2 \rfloor : \lceil k/2 \rceil$ and selects the ratio minimising $EC_{\text{norm}} = EC / \sqrt{\min(i, k - i)}$.

The $\sqrt{\min(i, k - i)}$ denominator is the isoperimetric correction: for a convex region enclosing population fraction i/k , the minimum boundary length scales as $\sqrt{\min(i, k - i)/k}$. This makes all ratios compete on equal footing: the $1 : (k - 1)$ ratio must win the isoperimetric competition against the balanced split, not merely be shorter in absolute terms.

EC_{norm} definition. Formally, $EC_{\text{norm}}(i) = EC(i) / \sqrt{\min(i, k - i)}$, where $EC(i)$ is the total TIGER-weighted boundary length of the cut edges produced by the $i : (k - i)$ first-bisection ratio and k is the total district count for the state. This normalisation penalises extreme ratios ($1 : (k - 1)$) relative to balanced ratios ($\lfloor k/2 \rfloor : \lceil k/2 \rceil$): a peel that isolates one district from the rest must clear a compactness bar $\sqrt{k/2 - 1}$ lower than the balanced split, preventing caterpillar districts in which the bisection tree degenerates into a spine of single-district peels. For n -way direct partitioning (as in the METIS k -way call used at the leaf level), the normalisation simplifies to $EC_{\text{norm}} = EC / \sqrt{k/2}$.

Key finding (T.1): GeoSection selects the balanced split (or near-equal split within 20%) for 12 of 44 multi-district states. For 32 states, it selects asymmetric ratios—but only where the urban core passes the isoperimetric test. North Carolina ($k = 14$) is the most important case: without normalisation, the Charlotte/Raleigh peel ($1 : 13$) wins; with normalisation, the east/west bisection ($6 : 8$) wins, reflecting the Piedmont corridor’s actual geographic structure.

C3: AreaSection — Dual-Constraint Ratio Scan

AreaSection (T.2) combines the GeoSection ratio scan with the dual-constraint vertex weights of A2. The Lorenz-curve pre-filter eliminates ratios where simultaneous population-and-area balance is geometrically infeasible before running any METIS calls.

Key finding (T.2): AreaSection shifts the natural ratio from asymmetric to near-equal in many densely urban states: Illinois moves from 1:17 to 8:9, New York from 1:16 to 13:13, Florida from 1:27 to 14:14. States with genuinely concentrated populations (Georgia 3:11, Massachusetts 2:7) retain asymmetric splits because the area constraint cannot be satisfied at more balanced ratios.

C4: ApportionRegions — Prime Factorisation Tree

ApportionRegions (T.4, planned) derives the bisection tree structure from the prime factorisation of k , linking it to the Huntington-Hill apportionment sequence. For $k = 8 = 2^3$, the tree is a perfect three-level binary tree. For $k = 14 = 2 \times 7$, the first split is $7 : 7$, then each half splits recursively using the factorisation of 7 (prime, so near-equal splits all the way).

Auditability property: The bisection tree is entirely determined by k , the seed count, and the state’s tract adjacency graph. There are no “hidden” parameters. The plan manifest need only record k and the seed; the full tree follows deterministically from the published algorithm.

C5: NestSection — Compatible Factorisation Spine

NestSection (T.6) extends ApportionRegions to multi-chamber consistency. Given congressional (C), senate (S), and house (H) seat counts, NestSection computes $g = \gcd(C, S, H)$ and derives a *compatible factorisation spine* that all three chambers share.

Anti-gerrymandering property (T.6): A shared geographic hierarchy is an anti-gerrymandering mechanism by construction. A partisan actor who locks in a congressional spine cannot draw senate boundaries that contradict it, because the senate tree is geometrically derived from the same trunk sequence. Eleven states achieve score 0 (exact nesting): Oregon, Alabama, Montana, Wyoming, and others where $g = \min(C, S, H)$.

C6: StabilitySection — Cross-Census CSS Filter

StabilitySection (T.8) is not a division strategy per se, but a post-hoc filter that evaluates whether the GeoSection natural ratio is stable across census years. The Census Stability Score (CSS) is computed as a weighted combination of seat stability, ratio stability, and proportionality-gap similarity.

Key finding (T.8): 67 % of comparable states are stable at $\Delta f \leq 0.05$ across the 2000–2010 census decade. Iowa is the canonical unstable state: its GeoSection natural ratio shifted from 1 : 3 (asymmetric Des Moines peel) in 2000 to 2 : 2 (near-equal east/west split) in 2010, reflecting dramatic suburban population dispersal. States with fixed municipal boundaries (Milwaukee, Philadelphia, Chicago) are highly stable; states with rapidly expanding metro footprints (Phoenix, Las Vegas) are low-stability.

4 Composing Requirements: The Algorithm Matrix

4.1 The Requirements $\times$ Algorithm Matrix

Table 1 is the central reference of this synthesis. It maps every legal requirement from Section 2 to its primary algorithm component, the METIS implementation detail, and the primary B-series evidence paper.

Table 1: Legal requirements mapped to algorithm components. All modes share the population-balance hard constraint (R1: $\text{ubvec}[0] = 1.001$). The “component” column names the B-series component that primarily addresses each requirement. “Mutex” = Callais mutual exclusion enforced by `callais_preflight`.

Legal requirement	Legal basis	Algorithm component	com-	METIS detail	Evidence
R1: Population equality $\pm 0.5\%$	<i>Wesberry v. Sanders</i> (1964); Karcher (1983)	All modes: hard pop. constraint		$\text{ubvec}[0] = 1.001$	B.1, B.7
R2: Contiguity	Art. I const. floor; all state constitutions	All modes: -contig flag		METIS METIS_PartGraphRecursive	B.1, T.1
R3: Compactness	Most state constitutions; <i>Shaw v. Reno</i> “bizarre shape”	Geographic edge weights (B.2) + GeoSection isoperimetric norm (T.1)		Edge weight = TIGER boundary length; $\text{EC}_{\text{norm}} = \frac{\text{EC}}{\sqrt{\min(i, k-i)}}$	B.2, T.1
R4: County/subdivision preservation	<i>Rucho v. Common Cause</i> ; <i>Harper v. Hall</i> ; most state re-districting statutes	County-sticky (T.3)	α	$w'(u, v) = (1 + \alpha) \cdot w_{\text{geo}}$ for same-county edges	T.3
R5: VRA §2 / Gingles Prong 1	52 U.S.C. §10301; Gingles (1986); <i>Allen v. Milligan</i> (2023); <i>Louisiana v. Callais</i> (2026)	VRASection alignment score (T.7)	align-	$A = \text{MVAP}_L - \text{MVAP}_R $; $w_{\text{vra}} = 0.40$; Mutex enforced	T.7
R6: No racial predominance	<i>Shaw v. Reno</i> (1993); <i>Miller v. Johnson</i> (1995); <i>Louisiana v. Callais</i> p. 36	VRASection $w_{\text{vra}} = 0.40 < \text{Shaw threshold}$; Callais p. 36 mutex enforced		Compactness weighted $1.5\times$ over alignment; no racial target	T.7
R7: Districting	<i>U.S. v. Davis</i> (2024)	VRASection alignment score (T.7)		$A = \text{MVAP}_L - \text{MVAP}_R $; $w_{\text{vra}} = 0.40$; Mutex enforced	T.7

4.2 Configuration Recommendations

No single configuration satisfies all eleven requirements simultaneously; the requirements are logically independent and sometimes conflicting. Table 2 lists the configurations recommended for specific legal contexts.

Table 2: Recommended configurations by legal context. Each row is an independent configuration choice — the columns are the three axes (A, B, C) from Section 3. *Key:* A1=pop-only, A2=pop+area, A3=VRA-alignment, A4=prop-partisan; B1=unweighted, B2=geo-edges, B3=county-sticky; C1=balanced, C2=GeoSection, C3=AreaSection, C4=ApportionRegions, C5=NestSection.

Legal context	Axis A	Axis B	Axis C
<i>Default — no specific legal claim</i>			
Baseline (B.1 standard)	A1	B1	C1
Compactness-optimised	A1	B2	C2
<i>VRA Section 2 claim</i>			
Gingles Prong 1 analysis	A3	B2	C2
VRA + county preservation	A3	B3	C2
<i>State-court partisan neutrality claim</i>			
Rodden-null proof	A2	B2	C3
County-fair + geographic balance	A2	B3	C3
<i>Auditability / expert witness</i>			
Auditable + HH-linked	A1	B2	C4
<i>Multi-chamber anti-stacking</i>			
Nested all chambers	A1	B2	C5
<i>Census stability litigation argument</i>			
Stability-filtered plans	A1	B2	C2 + CSS filter

4.3 The Mutual Exclusion Constraint

Rows containing A3 (VRASession) and A4 (ProportionalSection) are mutually exclusive. This is not a software limitation but a constitutional requirement under *Louisiana v. Callais* p. 36. The `callais_preflight` function in the `bisect` CLI enforces this at runtime:

```
bisect state --state AL --partition-mode vra-section \
  --proportional-section # ERROR
[BOUNDARY] callais_preflight: VRASession and ProportionalSection
cannot be combined in a single run (Callais p.36, disentanglement
requirement). Run each mode independently and compare results.
See docs/legal/FAIRNESS_DOCTRINE.md.
```

A court reviewing the plan manifest can verify compliance by checking that `partition_mode` is either `vra-section` or `proportional-section` but never both. The `whatif-manifest v1` JSON records the exact mode, parameter values, and SHA-256 hash of the input adjacency graph.

4.4 Combining Axes: Compatibility Rules

Within the mutex constraint, axes can be combined freely. The following table documents known compatibility and incompatibility:

Table 3: Axis combination compatibility. “OK” = fully compatible; “Soft” = compatible but computationally expensive or legally redundant; “Mutex” = mutually exclusive (Callais constraint or algorithmic conflict).

	A3 (VRA)	A4 (Partisan)	B3 (County-sticky)
A2 (Pop+Area)	OK (different axes)	Mutex (A3 vs A4)	OK
A3 (VRA)	—	Mutex (Callais p.36)	OK
A4 (Partisan)	Mutex (Callais p.36)	—	OK
C4 (ApportionRegions)	OK	OK	Soft
C5 (NestSection)	OK	OK	OK

5 The Bakeoff: Eight Combinations Across Four States

Evidence status and relationship to the governed national bakeoff. This May 2026 research table is not the later NRS v0.3 national bakeoff. It mixes confirmed single-profile results with explicitly estimated and pending cells, uses tract-level political and demographic inputs, and does not provide a common-block national comparison. Accordingly, cross-configuration patterns below are exploratory hypotheses wherever they depend on a marked cell. The governed NRS v0.3 package compares only NRS with official CD118 assignments on a frozen 2020 block universe and does not validate this eight-configuration matrix.

Prospective procedural extension. The original Wisconsin and national v1 protocols remain frozen. Runs precommitted on or after 2026-09-09 also apply `certified-ceremony-bakeoff-extension`. That extension preserves the map and computational outcomes below and adds a separate procedural evidence table: procedure class, $k - 1$ required cuts, certified-cut coverage, released depths, prefix verification, final-tree binding, witness policy, review delay, status and halt reason, transcript bytes, verification time, and frozen tamper- suite result. Unsupported or halted arms remain in the matrix.

The two scoreboards cannot be collapsed into a post hoc weighted winner. Proof strength does not imply desirable map outcomes, and assignment reproduction does not imply model-scoped optimality or staged public disclosure. No bakeoff reported in this section was run as an externally witnessed ceremony; those comparative procedural cells are pending.

A bounded path-8 fixture exercises the new scoreboard without changing that claim boundary. Its certified four-district arm completes three of three required cut certificates in two depth releases, verifies as a prefix, and binds the final tree. The matched METIS replay arm is classified as **replay-only**, with zero certified cuts and **unsupported** ceremony status. An independent script regenerates the categorical scoreboard and rejects a timestamp mutation. The fixture uses test receipts rather than validated external witnesses and does not rank map quality or execute the full production tamper corpus.

5.1 Preregistered Wisconsin Structure Proof Slice

On 2026-08-09, before inspecting comparative results, we froze a separate common-input proof slice on Wisconsin’s 2020 congressional tract graph. The slice holds 1,542 tracts, eight districts, total population, geographic edge weights, single outer search, and requested seed 0 fixed while varying four implemented structure families. Its package regenerated exactly after canonical district relabelling. Table 4 reports the validity gate, not a ranking.

Table 4: Post-remediation Wisconsin structure validation. The archived initial run retained two audit failures; all four repaired executions pass.

Structure	Final seed	Audit	Failure witness
Standard bisection	2	pass	none
Ratio-optimal	0	pass	none
Ratio-optimal-area	0	pass	none
Prime-factor	0	pass	none

The standard run documents an automatic balance retry from requested seed 0 to accepted seed 2. The standard and ratio-optimal assignments agree on 837 of 1,542 tracts (54.2802 %) after maximum-overlap district-label matching. Across all six post-remediation pairs, agreement ranges from 54.0208 % to 66.2776 %. This establishes reproducible assignment difference on this input, not a causal effect or a preferred plan.

The first governed run is preserved separately. In that run, ratio-optimal-area disconnected a district, while prime-factor disconnected a district and produced 0.704852 % population deviation against the advertised 0.5 % tolerance. The prime-factor manifest nevertheless marked population balance valid. Inspection found that ApportionRegions used a hidden 3 % final validation tolerance, the population flag was hard-coded, and the two affected METIS paths did not consistently use the contiguity-capable k-way routine. Final validation now uses the configured tolerance, the flag is derived from the audit certificate, and k-way contiguity options are enabled. The repaired four-pass package and its exact regeneration close the one-State implementation gate; they do not authorize national expansion without a newly frozen schedule.

The governed evidence is in `docs/experiments/neutral-algorithm-family-bakeoff-wi-2020/` under protocol `neutral-algorithm-family-bakeoff-v1`. Because this slice uses a different metric and input contract, it does not replace any estimated or pending cell in the older tables below.

5.2 Frozen National Structure Pilot

A second protocol, `neutral-algorithm-family-national-bakeoff-v1`, freezes the national multi-district cohort, a staged schedule, failure policy, and claim boundary before expansion. Its pilot orders RI, NE, CT, KY, SC, WI, AZ, and CA, spanning two through 52 congressional districts. Each State holds its 2020 tract graph, seat count, total population, geographic edge weights, single outer search, and requested seed 0 fixed across the same four structure families. The complete target is the 44 States with at least two House seats; the six at-large States are design exclusions rather than passes.

The pilot is an engineering gate selected to exercise increasing scale, prime and composite seat counts, the Wisconsin anchor, and the maximum-size State. It is not a probability sample. A passing pilot can establish bounded multi-State implementation validity and authorize the already frozen full phase; it cannot estimate national prevalence, rank the structures, or support geometric, electoral, demographic, VRA, causal, or legal conclusions. Only the complete 44-State phase could establish coverage of that frozen cohort, and it still could not identify a generally best algorithm.

The retained initial pilot attempt passed 27 of 32 cells and six of eight States. It exposed a recursive contiguity defect in equal non-NRS splits: an early recursive-METIS split could fragment a child region before a later contiguity-constrained call. After routing all non-NRS splits through k-way METIS with contiguity and minimum-connectivity options, focused Arizona and California probes passed and the clean governed matrix passed all eight States and all 32 cells. Across its 48 within-State structure pairs, maximum-overlap assignment agreement ranges from 49.0634 % to 100 %; alternative edge-cut ratios to standard-bisect range from 0.856479 to 5.441826. These are descriptive mechanics, not pooled estimates. An independent verifier reran all 32 cells and passed byte-identical normalized comparison.

The authorized full phase subsequently executed all 44 multi-district States and passed all 176 cells without a seed retry. Across 264 within-State pairs, maximum-overlap assignment agreement ranges from 47.6257 % to 100 %; the 132 alternative edge-cut ratios to standard-bisect range from 0.623301 to 5.441826. An independent execution reran all 176 native cells and regenerated the normalized evidence exactly. Edge cuts are reported to five decimal places, while canonical assignments are byte-compared. Thus the package establishes complete coverage of the frozen cohort. These descriptive mechanics cannot support a generally best algorithm or political, geometric, demographic, VRA, causal, or legal conclusions.

5.3 Experimental Design

We evaluate eight algorithm combinations on four states that concentrate the full range of redistricting legal challenges. Wisconsin (8D, 50.3 % D vote) is the canonical competitive state with a compact Milwaukee metro core. North Carolina (14D, 49.3 % D vote) has the Piedmont corridor geographic structure, a history of VRA litigation (*Thornburg v. Gingles*, 1986), and an active state-court partisan neutrality standard. Georgia (14D, 50.1 % D vote) has a substantial Black minority (33 % of VAP), significant VRA exposure after the Stacey Abrams-era map litigation, and rapid Sun Belt population growth. Alabama (7D, 37.1 % D vote) is the *Allen v. Milligan* (2023) state: the Supreme Court required two Black-opportunity districts from a state with 27.5 % Black VAP.

Eight combinations.

1. **Baseline:** Standard balanced bisection, unweighted edges (B.1 standard).
2. **Geo+GeoSection:** Geographic edge weights + GeoSection ratio scan (B.2 $\times$ T.1 compactness baseline).
3. **Geo+AreaSection:** Geographic edges + dual-constraint area balance (B.2 $\times$ T.2 geographic fairness).
4. **County+GeoSection:** County-sticky edges + GeoSection (T.3 $\times$ T.1 county-preserving).
5. **Geo+VRASection:** Geographic edges + VRA alignment score (B.2 $\times$ T.7 VRA-compliant), $w_{\text{vra}} = 0.40$.
6. **Geo+ApportionRegions:** Geographic edges + prime-factorisation tree (B.2 $\times$ T.4 auditable).
7. **County+AreaSection:** County-sticky + geographic area balance (T.3 $\times$ T.2 county + geographic balance).
8. **Geo+NestSection:** Geographic + multi-chamber spine (B.2 $\times$ T.6, state-specific; applicable where chamber counts allow).

Metrics. For each combination $\times$ state, we measure: (a) normalised edge cut (EC_{norm} , km) as a compactness proxy; (b) maximum population deviation (%); (c) D seat count and proportionality gap (pp); (d) majority-minority district count (CVAP $\geq 50\%$ Black or Hispanic); (e) first-level natural bisection ratio; (f) Lorenz p^* (population in densest 50% of land); (g) county split count (counties with tracts in two or more districts).

All partisan results use 2020 presidential two-party vote share (VEST/Fekrazad precinct-to-tract interpolation [Fekrazad, 2021]). All population results use 2020 Census P.L. 94-171 redistricting data.

5.4 Results Tables

Table 5 through Table 8 report the bakeoff results for Wisconsin, North Carolina, Georgia, and Alabama respectively.

Value provenance. Values without superscript are *confirmed empirical results* from T.1–T.8 experimental runs on 2020 Census tract data. Values marked † ($\ddagger$) are *estimates* derived from: (a) the theoretical relationship between edge-cut and proportionality gap established in T.1–T.2, (b) interpolation from confirmed values in adjacent configurations, or (c) model-derived predictions; each estimate carries an implicit uncertainty of approximately ± 1 seat and ± 3 pp for the proportionality gap. Values marked ‡ ($\ddagger$) are *pending*: they require the T.4 (ApportionRegions) or T.5 (ProportionalSection) implementations to run; they should not be cited as empirical results in litigation or policy contexts until the relevant paper’s experimental runs are complete.

Table 5: Bakeoff results: Wisconsin ($k = 8$, 50.3% D vote, 2020 Census). Goal: reduce -12.8 pp proportionality gap; maintain Milwaukee metro integrity. EC = normalised edge cut (km); Pop. dev. = maximum district population deviation (%); MM = majority-minority districts. County splits: WI has 72 counties; target ≤ 8 splits under county-sticky mode.

Configuration	EC (km)	Pop. dev.	D seats	Gap (pp)	MM	Ratio	County splits
1. Unweighted+Bisect	570 [†]	0.47 %	4	-0.3	0	4 : 4	14 [†]
2. Geo+GeoSection	90	0.48 %	3	-12.8	0	1 : 7	10 [†]
3. Geo+AreaSection	622	0.48 %	3	-12.8	0	3 : 5	9 [†]
4. County+GeoSection	297	0.48 %	3 [†]	$-12.8^{\dagger}$	0	3 : 5	6 [†]
5. Geo+VRASection	265 [†]	0.49 %	3 [†]	$-12.8^{\dagger}$	0	1 : 7 [†]	10 [†]
6. Geo+ApportionRegions	95 [‡]	0.48 %	3 [‡]	$-12.8^{\ddagger}$	0	1 : 7 [‡]	10 [‡]
7. County+AreaSection	912	0.49 %	3 [†]	$-12.8^{\dagger}$	0	4 : 4	5 [†]
8. Geo+NestSection	295 [‡]	0.48 %	3 [‡]	$-12.8^{\ddagger}$	0	4 : 4 [‡]	9 [‡]

Key finding: Config. 1 (unweighted) gives **4D/4R** (-0.3 pp) — most proportional.

All compactness-optimised configs (2–8) give 3D/5R (-12.8 pp): Milwaukee peel concentrates Dem voters.

Note: County+AreaSection (Config. 7) achieves 4:4 ratio at the cost of 912 km boundary ($10\times$ longer).

5.5 Cross-State Patterns

Pattern 0: The Compactness-Proportionality Paradox

The most striking exploratory pattern: the most compact algorithm (GeoSection, minimising boundary length) is *not* the most proportional. In Wisconsin, unweighted bisection produces **4D/4R** (-0.3 pp) — nearly perfectly proportional — while GeoSection produces 3D/5R (-12.8 pp). In Alabama, unweighted produces 1D/6R (-22.8 pp) while GeoSection produces 0D/7R (-37.1 pp). In Georgia, GeoSection and unweighted both give 6D/8R — but AreaSection (which maximises geographic balance) gives 5D/9R (-14.4 pp), *worse* than both.

These reversals are a direct consequence of Rodden’s geographic-sorting thesis [Rodden, 2019]: the optimal boundary from a compactness perspective passes through the suburban ring around a city, which by design concentrates urban Democratic voters into one compact district and disperses suburban voters into multiple Republican-leaning districts. Compactness and proportionality are not aligned objectives. A practitioner who claims “we used the most compact algorithm” cannot simultaneously claim “therefore the map is fair.”

Pattern 1: Geographic Sorting Dominates

Within this mixed confirmed/estimated table, geographic voter sorting is a plausible contributor to the proportionality gaps, while algorithm choice also changes several reported outcomes. In Wisconsin, GeoSection configurations (2–8) all produce 3D/5R; only the unweighted baseline (1) achieves 4D/4R by *avoiding* compactness optimization. In North Carolina, GeoSection produces **7D/7R** ($+0.7$ pp) — the Charlotte peel happens to create 7 roughly-balanced districts — while unweighted gives 5D/9R. In Alabama, 0D/7R is stable

Table 6: Bakeoff results: North Carolina ($k = 14$, 49.3% D vote, 2020 Census). Goal: prevent caterpillar pathology; determine whether VRASection changes MM outcomes. NC Black VAP 23.0%; Gingles conditions met for at least 2 MM districts.

Configuration	EC (km)	Pop. dev.	D seats	Gap (pp)	MM	Ratio	County splits
1. Unweighted+Bisect	580 [†]	0.48 %	5	−13.6	1	7 : 7	18 [†]
2. Geo+GeoSection	220	0.48 %	7	+0.7	1	6 : 8	16 [†]
3. Geo+AreaSection	284	0.49 %	6	−6.5	1	6 : 8	15 [†]
4. County+GeoSection	255	0.48 %	6 [†]	−6.5 [†]	1	6 : 8	11 [†]
5. Geo+VRASection	240 [†]	0.49 %	5 [†]	−13.6 [†]	2	1 : 13 [†]	16 [†]
6. Geo+ApportionRegions	225 [‡]	0.48 %	6 [‡]	−6.5 [‡]	1	7 : 7 [‡]	16 [‡]
7. County+AreaSection	299	0.49 %	6 [†]	−6.5 [†]	1	6 : 8	10 [†]
8. Geo+NestSection	230 [‡]	0.48 %	6 [‡]	−6.5 [‡]	1	6 : 8 [‡]	15 [‡]

Key finding: Config. 2 (Geo+GeoSection) gives **7D/7R** (+0.7 pp) — near-proportional.

Charlotte/Raleigh peel (6:8 ratio) naturally produces near-equal partisan split in NC.

Note: Config. 5 VRASection shifts to 1 : 13 (more peeling) — correct for Gingles but reduces D seats.

across all configurations except VRASection (which produces 1D/6R by achieving 2 MM districts).

These rows motivate, but do not identify, the relative contributions of geographic sorting and algorithm choice. Doing so requires completed cells, multiple seeds or converged ensembles, frozen election aggregation, and a design capable of separating those mechanisms.

Algorithmic neutrality and outcome neutrality. Algorithm neutrality means no partisan data enters the algorithm — it does not mean partisan outcomes are identical across algorithms. The table illustrates the testable possibility that partisan-data-free algorithms produce different partisan outcomes as bisection structure changes. It does not show that a partisan outcome is determined by geography alone once an algorithm is fixed: profile selection, resolution, weights, seeds, repairs, and legally required modifications remain consequential choices. Courts evaluating algorithmic maps under state partisan-neutrality standards (Pennsylvania’s “free and equal” elections clause; North Carolina’s *Harper v. Hall* proportionality doctrine; New York’s *Harkenrider v. Hochul* efficiency-gap standard) should therefore evaluate the *choice of algorithm* as the locus of potential bias, not the individual map produced by a given algorithm run. Whether a process or outcome satisfies a State standard is a legal question outside this table. A completed preregistered bakeoff could contribute counterfactual evidence, but the present estimated and pending cells cannot resolve either inquiry.

Pattern 2: Alabama VRASection Fixture

In the reported Alabama configurations, config. 5 (Geo+VRASection) is the only row with two majority-minority districts. That diagnostic does not establish the full *Gingles* conditions or what *Allen v. Milligan* requires for a particular plan. GeoSection’s 3 : 4 first-level split isolates Birmingham (Jefferson County) as one Black-majority district but leaves the

Table 7: Bakeoff results: Georgia ($k = 14$, 50.1 % D vote, 2020 Census). Goal: achieve 2 MM districts consistent with Allen v. Milligan; test geographic sorting. GA Black VAP 30.5 %, Hispanic VAP 10.4 %; Atlanta metro dominant. Estimated results; GA 2020 sweep was in progress at T.8 publication.

Configuration	EC (km)	Pop. dev.	D seats	Gap (pp)	MM	Ratio	County splits
1. Unweighted+Bisect	520 [†]	0.49 %	6	−7.3	1 [†]	7 : 7 [†]	20 [†]
2. Geo+GeoSection	100	0.48 %	6	−7.3	1	1 : 13	18 [†]
3. Geo+AreaSection	483	0.49 %	5	−14.4	1	3 : 11	17 [†]
4. County+GeoSection	127	0.48 %	6 [†]	−7.3 [†]	1	1 : 13	13 [†]
5. Geo+VRASection	120 [†]	0.49 %	5 [†]	−14.4 [†]	3 [†]	1 : 13 [†]	18 [†]
6. Geo+ApportionRegions	105 [‡]	0.48 %	6 [‡]	−7.3 [‡]	2 [‡]	7 : 7 [‡]	18 [‡]
7. County+AreaSection	648	0.49 %	5 [†]	−14.4 [†]	1	3 : 11	12 [†]
8. Geo+NestSection	110 [‡]	0.48 %	6 [‡]	−7.3 [‡]	2 [‡]	7 : 7 [‡]	17 [‡]

Key finding: AreaSection/County+AreaSection *reduce* D seats (5 vs 6) and worsen gap (−14.4 vs −7.3 pp). Geographic balance disperses Atlanta’s Dem voters; GeoSection’s 1:13 Atlanta peel is better for proportionality.
Note: VRASection 3 MM districts (estimated); ApportionRegions 2 MM districts (pending run).

Black Belt fragmented across the remaining subregion. VRASection’s 2 : 5 split concentrates the Black Belt, Montgomery, and Mobile coastal areas on one side, enabling two Black-opportunity districts downstream.

The fixture motivates testing whether its geographic subregion structure can support illustrative plans. It does not show that practitioners must use VRASection, that other methods cannot satisfy the first *Gingles* precondition, or that the resulting districts are legally compliant.

Pattern 3: County Preservation Trades Off Against Compactness

The marked county-sticky rows suggest higher edge cuts (+3–7 %) than their geographic-only counterparts, but several inputs to this comparison are estimated. This is the expected Pareto tradeoff: keeping counties intact requires cutting longer boundaries. The trade-off is most pronounced in Alabama (config. 7 vs. 2: +8.4 % EC increase), where county groupings do not align well with the geographic bisection structure. In Wisconsin, the trade-off is more modest (config. 4 vs. 2: +3.9 %), because Wisconsin’s counties are geographically compact and their boundaries partially align with the natural bisection directions.

Pattern 4: AreaSection Reduces Partisan Gap in Competitive States

AreaSection (config. 3) estimates a reduction in the proportionality gap in North Carolina (−13.6 pp under GeoSection to −6.3 pp under AreaSection) and Georgia (−14.4 pp to −7.2 pp). This is the expected effect of preventing urban peeling: by forcing geographic balance, AreaSection avoids creating maximally-efficient Republican geometries in suburban rings. The partisan direction is not systematic—AreaSection helps Democrats in NC/GA but is neutral in WI and AL.

Table 8: Bakeoff results: Alabama ($k = 7$, 37.1 % D vote, 2020 Census). Goal: achieve 2 Black-opportunity districts per Allen v. Milligan (2023). AL Black VAP 27.5 %; Black Belt in south-central, Jefferson County (Birmingham) in north. VRASession confirmed result: 2:5 first-level bisection; 2 MM districts confirmed (T.7).

Configuration	EC (km)	Pop. dev.	D seats	Gap (pp)	MM	Ratio
1. Unweighted+Bisect	530 [†]	0.47 %	1	−22.8	1	4 : 3
2. Geo+GeoSection	408	0.47 %	0	−37.1	1	3 : 4
3. Geo+AreaSection	534	0.48 %	0	−37.1	1	3 : 4
4. County+GeoSection	570	0.47 %	0 [†]	−37.1 [†]	1	3 : 4
5. Geo+VRASession	420 [†]	0.49 %	1 [†]	−22.8 [†]	2	1 : 6
6. Geo+ApportionRegions	1730 [‡]	0.47 %	0 [‡]	−37.1 [‡]	1 [‡]	1 : 6 [‡]
7. County+AreaSection	1870 [†]	0.48 %	0 [†]	−37.1 [†]	1 [†]	3 : 4 [†]
8. Geo+NestSection	1740 [‡]	0.47 %	0 [‡]	−37.1 [‡]	1 [‡]	3 : 4 [‡]

Note: In this fixture, VRASession is the only reported row with 2 MM districts; this is not an Allen v. Milligan

Note: VRASession 2 : 5 result confirmed empirically (T.7); 2 MM at 55.2 % and 52.8 % Black VAP.

Note: EC premium of +4.3 % over GeoSection is modest and legally defensible.

Pattern 5: NestSection and ApportionRegions Are Structurally Neutral

The pending values for configs. 6 and 8 (ApportionRegions and NestSection) were projected to be close to config. 2 (GeoSection). They cannot support an empirical similarity or neutrality finding until executed. They are designed as audit and consistency tools rather than partisan tools. Their value is not in changing partisan outcomes but in providing structure-from-first-principles: the bisection tree is derived from mathematical properties of k (prime factorisation) or from multi-chamber consistency requirements, not from any geographic or demographic optimisation.

5.6 Metric Summary

Table 9 provides a cross-configuration summary aggregated across all four states.

6 Callais Compliance Architecture

6.1 The Disentanglement Requirement

Louisiana v. Callais, 608 U.S. --- (2026), at p. 36, articulates the disentanglement requirement as follows:¹ a redistricting process that combines race-conscious and partisan signals in the same objective function is constitutionally suspect because “a court cannot isolate

¹All page citations to *Louisiana v. Callais* refer to the slip opinion as issued by the Court. The final U.S. Reports citation and permanent page numbers will be updated when the opinion is paginated in the bound volume. The disentanglement language quoted here is from the majority opinion; the slip opinion does not carry section headers distinguishing majority, concurrence, or dissent by page number.

Table 9: Cross-configuration metric summary, averaged across WI, NC, GA, AL. EC change relative to config. 2 (Geo+GeoSection = baseline). Gap change is arithmetic change in proportionality gap (pp). MM change is change in MM district count. County splits change relative to config. 2.

Configuration	EC change (%)	Gap change (pp)	MM change	County split change
1. Baseline	+8.3 [†]	−5.7 [†]	−0.3 [†]	+2.5 [†]
2. Geo+GeoSection (ref)	0	0	0	0
3. Geo+AreaSection	−0.8 [†]	+4.4 [†]	0 [†]	−1.3 [†]
4. County+GeoSection	+3.9 [†]	0 [†]	0	−4.8 [†]
5. Geo+VRASection	+3.1 [†]	+0.7 [†]	+1.0 [†]	0
6. Geo+ApportionRegions	+0.2 [‡]	0 [‡]	+0.3 [‡]	0 [‡]
7. County+AreaSection	+6.1 [†]	+1.1 [†]	0 [†]	−5.5 [†]
8. Geo+NestSection	+0.5 [‡]	0 [‡]	+0.3 [‡]	−1.0 [‡]

[†] = estimated from B-series models; [‡] = pending T.4/T.5 implementation.

the racial motivation from the partisan motivation in evaluating the resulting map’s constitutionality.” The majority did not hold that either signal is impermissible; it held that the two signals cannot be *combined* in a way that makes their individual contributions undecipherable.

This requirement maps precisely onto the toolbox architecture. The toolbox has exactly two signal-bearing modes:

- **VRASection (T.7):** Geographic alignment score on minority VAP. Permissible under *Louisiana v. Callais* because (a) it addresses a geographic property (where minority population is concentrated), (b) it enters only as a ratio selector, not as a METIS constraint, and (c) compactness is the predominant criterion at $w_{\text{vra}} = 0.40$.
- **ProportionalSection (T.5, planned):** Partisan vote-share balance constraint. Permissible under *Rucho v. Common Cause*’s state-court exception because it is a partisan signal, not a racial signal, and it addresses the proportionality criterion directly. Mutex-gated against VRASection.

6.2 How VRASection Avoids Racial Predominance

VRASection’s design makes three specific architectural choices to stay below the *Shaw v. Reno–Miller v. Johnson* racial-predominance threshold.

Choice 1: Geographic proxy, not racial count. The alignment score $A = |\text{MVAP}_{\text{frac}}(L) - \text{MVAP}_{\text{frac}}(R)|$ measures how much minority VAP is concentrated on one side of a proposed bisection. This is a *geographic property* of the existing population distribution, not a racial target. The algorithm does not specify “the left side must contain at least $X\%$ minority population”; it prefers bisections that *happen to place* a compact minority community on one side because that is what minimises the score function. The geographic structure drives the decision; the racial measurement merely *detects* that structure.

Choice 2: Compactness-dominated weight. At $w_{\text{vra}} = 0.40$, the score function is:

$$\text{score}(\text{ratio}) = \underbrace{\text{EC}_{\text{norm}}}_{\text{compactness}} - \underbrace{0.40 \cdot A \cdot \max(\text{EC}_{\text{norm}}, 1)}_{\text{alignment bonus}}$$

The alignment bonus can reduce the compactness penalty by at most $0.40 \times A$, where $A \leq 1$. The minimum alignment effect is zero; the maximum is a 40 % reduction in the compactness score. In practice, for Alabama ($A = 0.42$ at the 2 : 5 winning ratio), the alignment bonus reduces the score from 448.7 to 322.5—a 28 % reduction. The compactness penalty (448.7) is still the dominant term; the alignment bonus is a correction, not a driver.

This weight satisfies the *Miller v. Johnson* test as interpreted in *Bush v. Vera* (1996): “race was not the predominant factor overriding traditional districting principles.” Traditional compactness (as measured by normalised edge-cut) is the dominant criterion at $w_{\text{vra}} = 0.40$.

Choice 3: No MetisVra failure mode. The MetisVra approach [Della-Libera, 2026a] added minority VAP as a second METIS balance constraint. This fails in 3 of 5 majority-minority states due to numerical scale mismatch: minority VAP fractions (0.0–1.0) and census population counts (10,000–200,000) differ by 60–800×. More fundamentally, MetisVra *racially targets*: it specifies an explicit minority fraction that the district must achieve. VRASection avoids this by never specifying a target minority fraction. The alignment score measures geographic concentration; it does not mandate a racial outcome.

6.3 The CLI Mutex Gate

The `callais_preflight` function is the runtime enforcement mechanism. It is called at the start of every `bisect state` and `redist run` invocation, before any data is loaded:

```
fn callais_preflight(args: &StateArgs) -> Result<()> {
    if args.partition_mode == VraSection
        && args.proportional_section {
        return Err(CallaisViolation {
            message: "VRASection and ProportionalSection cannot be \
                    combined in one run (Callais p.36). Run each \
                    mode independently.",
            citation: "Louisiana v. Callais, 608 U.S. ___ (2026), p.36",
        });
    }
    Ok(())
}
```

The function also checks for weaker disentanglement violations: if the user sets $w_{\text{vra}} > 0$ while also specifying `--election-year` (which enables election-result analysis), a warning is emitted:

[BOUNDARY] callais_warning: VRASession with --election-year set.
Election results will be recorded in the manifest but were NOT
used as input to the partition algorithm. Confirm this is correct.

This warning is not a blocking error—election results can legitimately be recorded for post-hoc analysis—but it documents that the partisan data was not part of the optimisation.

6.4 The Plan Manifest as Audit Record

Every `bisect` run produces a `whatif-manifest v1` JSON sidecar with the following fields relevant to Callais compliance:

```
{
  "partition_mode": "vra-section",
  "w_vra": 0.40,
  "proportional_section": false,
  "election_year": null,
  "callais_preflight_passed": true,
  "callais_preflight_version": "2026-04-30",
  "inputs": {
    "adj_graph_sha256": "a3f7...",
    "minority_vap_source": "Census P4 table, 2020",
    "partisan_data_source": null
  },
  "overrides_hash": "d5e2...",
  "log_sha256_chain": ["...", "..."]
}
```

A reviewing court or opposing expert can verify: (a) the partition mode is `VRASession` but not `ProportionalSection`, (b) partisan data was not an input (`partisan_data_source: null`), (c) the `callais_preflight` check was passed (not bypassed), (d) the adjacency graph SHA-256 matches the publicly released data, (e) the overrides hash is consistent with the stated parameters, and (f) the log SHA-256 chain has not been tampered with (single-byte tamper is detectable).

6.5 The Strong-Inference Test Applied to the Toolbox

Louisiana v. Callais’s “strong-inference” framework (holding 2) requires §2 challengers to demonstrate that the state’s stated political goals could have been achieved *with better minority outcomes*. The toolbox directly provides this counterfactual analysis.

Procedure.

1. Run `GeoSection` (no minority signal) on the disputed state. Record D seat count and MM district count. This is the “pure political goals” baseline.

2. Run VRASession ($w_{\text{vra}} = 0.40$) on the same state. Record D seat count and MM district count.
3. If VRASession achieves at least as many D seats as GeoSection *and* more MM districts, the strong-inference test is satisfied: the state’s political goals could have been achieved with better minority outcomes.

For Alabama:

- GeoSection (stated political goal: efficient district layout): 0D, 1 MM district.
- VRASession: 1D, **2 MM districts** (confirmed, T.7).
- Research diagnostic: the fixture produces a second threshold-based minority-opportunity district. This does not adjudicate strong inference or establish an *Allen v. Milligan* remedy.

For Georgia:

- GeoSection: 5D, 1 MM district (estimated).
- VRASession: 5D, 3 MM districts (estimated).
- Strong-inference status: unavailable because both comparison rows are estimated.

For North Carolina:

- GeoSection: 5D, 1 MM district.
- VRASession: 5D, 2 MM districts (estimated, Charlotte/Raleigh isolation).
- Strong-inference status: unavailable because the VRASession row is estimated.

Strong-inference interpretation. The procedure above compares two distinct algorithm configurations (GeoSection, VRASession) and therefore conflates two potential sources of MM-count improvement: the ratio-selection mechanism (choosing a different first-level split) and the alignment signal ($w_{\text{vra}} > 0$). To isolate the alignment signal’s causal contribution, T.7 includes an ablation in which VRASession is run with $w_{\text{vra}} = 0$, which reduces to GeoSection in the ratio-selection step. The ablation isolates the alignment signal within the Alabama fixture: at $w_{\text{vra}} = 0$, Alabama produces 3 : 4 first-bisection and 1 MM district (identical to GeoSection); at $w_{\text{vra}} = 0.40$, Alabama produces 2 : 5 and 2 MM districts. This mechanism result does not identify causal partisan effects. The broader six-mode claim is not testable from this matrix because some algorithm rows are pending or estimated; comparison with within-algorithm seed variance likewise requires a completed common-input execution.

6.6 Separation of Legal Signals: A Formal Statement

Proposition 1 (Callais compliance of the toolbox). *The algorithmic redistricting toolbox satisfies the Louisiana v. Callais p. 36 disentanglement requirement if and only if the following conditions hold:*

1. *At most one of $\{\text{VRASection}, \text{ProportionalSection}\}$ is active in any single `bisect` run.*
2. *The `callais_preflight` gate is not bypassed (no `--skip-preflight` flag).*
3. *The plan manifest `whatif-manifest v1` is produced and its hash chain is unbroken.*

Under these conditions, a reviewing court can determine from the manifest alone which signal was used, verify that no other signal was present as an input, and reproduce the run from the stated parameters.

Proof. Conditions 1–2 prevent the simultaneous use of race-conscious and partisan signals. The `callais_preflight` function is called before any data is loaded; it cannot be bypassed without modifying the binary. Condition 3 provides the audit record: the SHA-256 hash of the adjacency graph verifies the data provenance; the `overrides_hash` verifies the parameters; the log hash chain verifies post-hoc integrity. A court can reproduce the run by providing the adjacency graph (hash-verified) and the stated parameters, and checking that the resulting partition matches the produced map. $\square$

7 Discussion

7.1 Which Configuration for Which Use Case

The requirements matrix identifies configurations for further evaluation; the mixed confirmed, estimated, and pending bakeoff does not determine which configuration a practitioner should use. We organise research candidates by use case, subject to completed execution and independent legal review.

Default (no specific legal claim). Use **Geo+GeoSection** (config. 2). It satisfies population equality, contiguity, and compactness (56% improvement over baseline), does not manipulate county lines or racial geography, and is the best-documented configuration in the B-series. The plan manifest is clean: only geographic signals are present.

Allen v. Milligan exposure (VRA Section 2, Gingles Prong 1). Evaluate **Geo+VRASection** (config. 5), $w_{\text{vra}} = 0.40$, as one illustrative-plan candidate. The Alabama fixture reports two threshold-based minority-opportunity districts; the Georgia cells are partly estimated. Neither the reported compactness cost nor the parameter value establishes VRA compliance or satisfies the *Miller v. Johnson* racial-predominance inquiry.

State-court partisan neutrality claim (LWV v. PA; Harper v. Hall). Evaluate **Geo+AreaSection** (config. 3) as a geographic-balance sensitivity diagnostic. It cannot prove that a map is not a product of algorithmic partisan bias. **AreaSection** reduces the proportionality gap in NC and GA by 7–8pp in our estimates, not because the algorithm adds partisan signals but because it prevents the urban-peeling geometry that creates artificially efficient Republican configurations. If the state constitution requires proportionality, **ProportionalSection** (T.5, when implemented) is the direct tool—but it cannot be combined with **VRASection** in the same run.

County preservation as a statutory criterion. Use **County+GeoSection** (config. 4) with α_{county} tuned to the state’s statutory standard. For states like Wisconsin (strong county preservation tradition) or California (explicit statutory criterion), a higher α reduces county splits at a modest EC cost. The estimated rows suggest a possible county-split/edge-cut tradeoff; a completed common-input sweep is required to quantify it.

Auditability and expert witness. Treat **Geo+ApportionRegions** (config. 6) as pending empirical evaluation when the primary concern is reproducibility and parameter-freedom. The bisection tree is derived entirely from the prime factorisation of k ; there are no “hidden” parameter choices about which ratio to select. Combined with the plan manifest, this provides the cleanest audit record.

Multi-chamber anti-stacking. Use **Geo+NestSection** (config. 8) in states where all three chambers are drawn simultaneously and the **NestSection** compatibility score allows it. The eleven states with spine score 0 (exact nesting possible) are the best candidates; the remaining states require best-effort Mode 3 nesting.

7.2 Limitations

L1: Geographic Sorting Cannot Be Eliminated

Geographic voter sorting is a plausible source of systematic partisan asymmetry, but this paper does not estimate its effect independently of algorithm and profile choices. All configurations show a mean proportionality gap of -7 pp in the eight competitive states studied across T.1–T.2. **AreaSection** reduces this to approximately -4 pp in the best case by preventing the caterpillar pathology, but it cannot eliminate the underlying geographic effect. A state where Democrats are concentrated in three or four major cities and Republicans are dispersed across the rural remainder will produce more Republican seats than the vote-share would suggest under any geographically-neutral algorithm. This is a central mechanism studied by Rodden [2019]; the B-series results motivate a preregistered causal or ensemble study rather than independently confirming it across 44 States.

L2: VRASection Does Not Guarantee MM Districts

VRASection’s 2 : 5 first-level split in Alabama concentrates minority population in a geographic subregion; it does not guarantee that any specific district within that subregion

will be majority-minority. The recursive GeoSection within the southern subregion must still achieve $\geq 50\%$ Black VAP in at least two districts. T.7 confirms that this does occur (55.2% and 52.8% Black VAP in the two southern districts), but the guarantee is empirical, not structural. For states where the minority population is more dispersed, or where the 2 : 5 subregion has Black VAP just above 50%, the guarantee may not hold. Users should run `redist analyze --types demographic` to verify MM district counts after every VRA-Section run.

L3: Census Stability is Not Universal

StabilitySection (T.8) shows that 33% of states are *unstable*—their GeoSection natural ratio changes between census years. The census-stability litigation argument cannot be made for these states. Iowa is the canonical example: its natural ratio shifted from 1 : 3 (2000) to 2 : 2 (2010) as suburban population dispersal changed the population-geographic structure. For Iowa-class states, the practitioner should present the map as reflecting the *current census year’s geographic structure*, not a census-stable property.

L4: ProportionalSection is Planned, Not Implemented

The ProportionalSection (T.5) and ApportionRegions (T.4) components are described in this synthesis but not yet implemented in the `bisect` CLI. Results for configs. 6 and 8 in the bakeoff tables are pending. The mutual-exclusion architecture described in Section 6 is fully implemented and will enforce the ProportionalSection mutex as soon as T.5 is complete.

L5: County-Sticky Weights Require TIGER Download

Full subdivision-stickiness (MCD, place, VTD levels) requires downloading additional TIGER spatial files (~ 2 GB for all 50 states) and performing spatial joins. County-level stickiness is available immediately (county FIPS is encoded in the census-tract GEOID); finer levels require the T.3 Phase 2–3 implementation.

7.3 Comparison with Existing Approaches

Markov Chain Methods (GerryChain)

GerryChain [MGGG Redistricting Lab, 2018] and related Markov-chain Monte Carlo (MCMC) approaches generate large ensembles—typically thousands of independently drawn plans—and test whether an enacted map falls within the ensemble’s distribution. These methods are well-suited to null-hypothesis testing (“is the enacted map unusual relative to the space of plans?”). The toolbox is complementary: it generates a single deterministic plan optimising a specified criterion, while GerryChain generates an MCMC ensemble of thousands of plans to characterise the solution space. A practitioner may use the two approaches together: run the toolbox to obtain the certified-optimal plan, then use GerryChain to verify that the plan lies in the interior of the ensemble (i.e., is not an outlier in any partisan or demographic dimension). T.8’s Census Stability Score provides a deterministic stability metric that MCMC cannot easily compute (it would require running an ensemble for each of three census years).

Integer Programming

Integer programming (IP) approaches (e.g., Gurnee and Shmoys [2021]) can optimise multiple objectives simultaneously but do not scale to large states. For California (52 districts, $\sim 8,000$ census tracts), IP is intractable. The METIS-based toolbox operates in $O(n \log k)$ time and produces results in 30–90 seconds per state on a single CPU. The trade-off is that METIS produces locally optimal (not globally optimal) solutions; the B.7 seed study shows this produces a narrow solution space with < 0.3 seats of variation per state.

Short Burst Methods

Short burst methods [Cannon et al., 2023] run many short Markov chains to explore the redistricting solution space. They are well-suited to understanding solution space geometry but share the MCMC limitation that they are not deterministic: two independent runs of the same method on the same state may produce different plans. The toolbox’s auditable determinism (plan manifest + hash chain) is a structural advantage for litigation, where reproducibility is a requirement, not a preference.

7.4 The Algorithm as Evidence, Not Advocacy

A final point deserves emphasis. The toolbox is designed as an *evidentiary* tool, not an advocacy tool. It does not argue for a particular redistricting outcome. It provides a set of analytically defensible configurations, each grounded in a specific legal requirement, each producing a reproducible and auditable result.

A redistricting commission, a court-appointed special master, or an opposing expert witness can all use the same toolbox and reach different conclusions—because the conclusions depend on which legal requirements are weighted most heavily, not on which algorithm is used. VRASection and GeoSection will sometimes produce different seat counts (North Carolina, Wisconsin) and sometimes not (Alabama with 0D in both). The algorithm does not choose; the legal argument and the factual record choose. The toolbox provides the counterfactual analysis that makes the argument *auditable*.

8 Conclusion

8.1 Summary

We have presented a candidate algorithmic toolbox and requirements map for congressional-redistricting research. The implementation and evidence described here do not by themselves satisfy the eleven legal requirements or establish a post-*Louisiana v. Callais* compliance framework.

The toolbox is organised along three orthogonal axes: Axis A controls what METIS balances (population only, population plus area, population with VRA alignment, population with partisan vote share); Axis B controls edge management (geographic boundary lengths, county-sticky weights, unweighted); Axis C controls division strategy (balanced bisection, GeoSection, AreaSection, ApportionRegions, NestSection).

Each legal requirement maps to at most one component on one axis. The components compose freely except for the Callais mutex: VRASection (Axis A3) and ProportionalSection (Axis A4) are mutually exclusive, enforced at the CLI level by `callais_preflight` and documented in the plan manifest.

The four-State bakeoff retains confirmed, estimated, and pending cells. It therefore generates hypotheses about geographic sorting, algorithm choice, minority-opportunity diagnostics, subdivision tradeoffs, and audit structure; it does not establish cross-State effects, causal mechanisms, legal compliance, or algorithm rankings. The separately preregistered Wisconsin proof slice establishes a narrower implementation history. Its initial two validity failures exposed hidden tolerance, reporting, and contiguity-path defects. After remediation, all four plans passed their native audits and regenerated exactly. The subsequently frozen eight-State national pilot retained a second failure witness, exposed an equal-split recursive contiguity defect, and passed and exactly regenerated all 32 cells after remediation. The pilot cannot support national prevalence, causal effects, or a ranking, and the frozen 44-State clean matrix subsequently passed all 176 cells without seed retries, and an independent 176-cell execution regenerated the normalized evidence exactly. Its scope is implementation validity rather than comparative superiority or legal adequacy.

8.2 The Callais Compliance Design

Louisiana v. Callais's disentanglement requirement at p. 36 is the structurally decisive constraint for algorithmic redistricting. The proposed toolbox is designed to separate signal types: each mode uses exactly one signal type, the two signal-bearing modes are mutex-gated, and every run is documented in a tamper-detectable plan manifest.

The strong-inference framework of *Louisiana v. Callais* holding 2 is directly supported: a practitioner can run GeoSection (pure political goals) and VRASection (minority alignment) independently, compare the results, and demonstrate that the political goals could have been achieved with different threshold-based minority outcomes. For Alabama, this is an illustrative research comparison, not confirmation of an *Allen v. Milligan* remedy.

8.3 The Central Empirical Finding

The central hypothesis is that geographic voter sorting materially contributes to partisan redistricting outcomes. This matrix does not estimate that effect separately from algorithm, profile, resolution, seed, and repair choices, and it does not support a claim that outcomes are invariant to district lines.

This finding has a constructive implication. The evidence here cannot adjudicate a claim that an algorithm favoured a party. A relevant empirical question is whether another valid, comparably constrained method could achieve the same stated goals with different minority outcomes, but answering it requires a completed counterfactual design rather than this mixed-evidence matrix or the one-State mechanics slice. For *Allen v. Milligan* matters, VRASection may help construct an illustrative diagnostic; it does not provide the legal or factual proof required by a court.

8.4 Future Work

Four extensions are needed to complete the toolbox:

1. **T.4 ApportionRegions**: Extend the completed, exactly regenerated frozen-cohort validity result only under separately preregistered comparative designs, preserving every failure and separating validity from performance.
2. **T.5 ProportionalSection**: Implementation with the Callais-compliant mutex gate; empirical comparison against AreaSection in state-court proportionality challenge states.
3. **T.3 Phase 2–3**: Download and spatial join of TIGER MCD, place, and VTD files; parameter sweep at the township and precinct levels for MN, WI, and New England states.
4. **T.8 Three-Census CSS**: Completion of the 2010 GeoSection sweep to compute full three-census CSS for all 50 states.

These extensions are prerequisites for a broader auditable research toolbox; they would not by themselves establish a legally sufficient framework for a future redistricting cycle.

Data Availability. The `bisect` Rust binary implementing GeoSection, VRASection, AreaSection, NestSection, and StabilitySection is available as open-source software at <https://github.com/apportionment-research/redist>. All redistricting outputs, adjacency graphs, and analysis scripts are available at the same repository. Pre-computed adjacency files for all 50 states are available as GitHub Release assets, with SHA-256 hashes recorded in the plan manifests. All parameters are documented in `docs/REDIST_CLI.md`. Marked empirical results are intended to be reproducible from their stated parameters and inputs. Estimated and pending cells are not execution results.

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